Linkability of LLM Digital Twins: From Chance in 2023 to 60 % in an Open World, and the Control That Says When Not to Measure

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Abstract

Published LLM "digital twins" — per-respondent simulated answer vectors released alongside survey panels — are linkable to the humans they were built from, and that risk has a trajectory. Measured on three datasets and three generations of models, re-identification was indistinguishable from chance with the twins of 2023 and reaches 60 % in an open world today. On the GPT-3 twins Argyle et al. released in 2023, the strongest twin designates the right respondent among 2,148 in **0.14 %** [0.05 ; 0.27] of cases against 0.10 % [0.02 ; 0.23] for a demographic baseline recomputed on that same pool — intervals that overlap — the best of the three carrying 0.094 bits of identity against a ceiling of 11.07, at a per-item accuracy of 47.2 %, *below* the modal answer; a nearest neighbour holding the same eleven true answers identifies 3.6 times better. On twins built with recent models that attack reaches **20.7 %** [19.0 ; 22.4] on Twin-2K-500 (2,058 respondents) and, in a rarity-weighted variant, **90.40 %** [88.6 ; 92.2] on the Park et al. archive (1,052 participants) in a closed pool, and in an open world at 1 % false accusations, **4.28 %** [3.3 ; 5.6] and **60.17 %** [54.5 ; 64.4], against human test-retest ceilings of 54.5 % and 90.7 %. **The three datasets differ in team, protocol, item count (12 against 60 and 177) and pool, so "the risk grew with model capability" is confounded with "the protocols differ"; we state that confound in front rather than resolve it.** We report a twin-to-twin channel: within one pipeline, two twins of the same person designate each other with no real human answer held by the attacker — 36.4 % top-1 at 60 common items on Twin-2K-500 and 11.9–13.0 % at 177 common items on the Park archive, against anti-artifact controls of 0.06 % and at most 0.31 %. What governs that channel is *which* pipeline element differs, not how many: at a single element changed, rates run from 17.2 % (model) to 81.4 % (decoding), and the "distance law" we preregistered is refuted by our own measurements. We contribute a control that costs nothing: no experimental arm is interpretable until its candidate pipeline beats a demographic baseline against the real humans. It stopped two of our own paid experiments and the Argyle replication. We also report what we could not establish. We predicted, and preregistered, that the observed coupling between imitation quality and leakage reflects an individual fingerprint; a null carrying only a matched per-person accuracy margin — not a structure-free object, and one that in fact leaks more than our own twin (31.6 % against 20.7 %) — reaches a rank correlation as high as ours (0.974, 5th–95th percentiles [0.950 ; 0.993], against 0.965 observed). Fourteen preregistered predictions were refuted, two are inconclusive for lack of power, and one could not be tested; our own twins failed to reproduce the leakage even with a paid frontier model. Finally, a 1998 mechanism (PRAM) applied to twins reduces closed-world top-1 from 20.7 % to 0.13 % at a measured cost of 4.4 points on inter-item correlations, holding at 0.29 % against an adaptive attacker who knows the mechanism.

Proceedings on Privacy Enhancing Technologies 2027(3), 1–15

https://doi.org/XXXXXXX.XXXXXXX

Keywords

digital twins, large language models, re-identification, linkability, synthetic survey data, privacy

1 Introduction

Survey research is beginning to publish *digital twins*: for each human respondent in a panel, an LLM is conditioned on that person's profile and its simulated answers to the questionnaire are released as a per-person record. Toubia et al. published 2,058 such twins as Twin-2K-500 [38]. Park et al. built generative agents for 1,052 participants and cited privacy in restricting access to individual agent responses [27] — although the publicly posted replication package does carry those individual responses together with the agent conditions we analyse (§ 4.1, Ethical Considerations). The privacy question these releases raise has been named — by Park et al. themselves, by Bonagiri et al. [4], and by the AAPOR task force [32], which ranks synthetic response generation as the most risky of the core tasks it evaluates and names re-identification by linkage without quantifying it — but it has not been measured.

This paper measures it on three datasets spanning three generations of models, and reports both what the measurement supports and what it refutes.

1.1 The trajectory, and the confound that comes with it

Argyle et al. published in 2023 GPT-3 twins of ANES respondents, 12 matched items per person [3]. Attacked exactly as we attack the recent panels, those twins designate the right person among 2,148 in **0.14 %** [0.05 ; 0.27] of cases, against **0.10 %** [0.02 ; 0.23] for a demographic baseline recomputed on that same pool: the intervals overlap. The best of the three carries 0.094 bits of identity against a ceiling of 11.07, per-item accuracy of 47.2 % sits *below* the modal answer, and a nearest neighbour given the same eleven true answers identifies **3.6 times better**. Three years later, twins of the same kind of panel are re-identified at 90.40 % in a closed pool and 60.17 % in an open world at 1 % false accusations. The risk was not measurable in the first published generation of twins; it is substantial in the current one.

The confound, stated before the claim rather than after it. The three datasets differ in team, protocol, item count (12 against 60 and 177), pool and questionnaire. "The risk grew with model capability" is therefore confounded with "the protocols differ", and nothing we measure separates them. Lifting the confound would require the same items put to the same population by several generations of models, with individually matched twins released for each; no such resource is public (§ 7.4). We state the trajectory as a measured contrast between three published corpora, never as a causal effect of model capability.

One reading must be excluded. The Argyle result is **not a refutation of our scaling law**: it is out of domain. That law predicts the leakage of a twin that carries a person, and our interpretability control (§ 4.5) establishes that these twins carry none we can demonstrate. Taking 0.14 % for a refutation would commit, in reverse, the exact error the control exists to prevent — concluding from a measurement taken on noise (§ 5.8).

1.2 What we set out to show, and what happened

Our central hypothesis was that a twin's fidelity to *its particular person* and that twin's identifiability are expressions of one underlying axis. We preregistered a self-refutation test in two parts. The first passed: the rank correlation between imitation quality and leakage survives measuring the two axes on completely disjoint item sets (Spearman 0.969, 95 % CI [0.937 ; 0.993], 50 stratified random A/B splits of 60 items). The relation is not an artifact of recycled computation.

The second part failed, and against us. We constructed a *marginal null*: 100 artificial predictors matched to each person's own accuracy margin, with correct positions drawn at random without looking at the person — an object that is not structure-free, and that in fact leaks more than our own twin (31.6 % top-1 against 20.7 %). Adversarial review found a real defect in that first witness, since rebuilt without changing the verdict (§ 5.1). The rebuilt witness reaches mean Spearman **0.974**, 5th–95th percentiles [0.950 ; 0.993], against **0.965** observed over the 12 configurations; the observed value does not exceed it. Each null is compared only against the observed value measured on its own plan. **Reproducing the coupling requires nothing more than a matched per-person accuracy and correct positions drawn at random — prediction (b) is refuted.**

We report this as a result rather than a confession. A recent line of work argues that an attack on foundation models proves nothing until the null hypothesis is properly sampled: blind baselines routinely beat state-of-the-art membership inference attacks [8, 11], and a SaTML position paper makes the demand explicit [42]. The quality-leakage coupling reported in prose by Ward et al. [39] and Byun et al. [6], and contested by Platzer and Reutterer [29] and Adams et al. [1], has never been tested against such a control. We built it, and it absorbs our own effect. What survives is not a theory of why twins leak, but a set of measurements that do not depend on the coupling being individual-specific at all.

1.3 Contributions

(1) A trajectory measured across three generations of published twins. The same attack, under the same control, on twins released in 2023, 2025 and 2026: chance-level then, 60.17 % in an open world now (§ 5.8) — with the protocol confound of § 1.1, which forbids reading it as an effect of model capability alone.

(2) An interpretability control, and what it cost us. A test that consumes no model call and no budget: **no experimental arm is interpretable until its candidate pipeline has been shown to beat the demographic baseline in top-1 against the real human answers** (§ 4.5). It stopped two of our own paid arms (§ 5.4), the Argyle replication (§ 5.8), and the `persona` condition of the Park archive — an agent built on a person's own self-description re-identifies her *less* well than demographics alone, 0.26 % [0.06 ; 0.52] against a baseline of 0.39 %. Any team measuring privacy on simulated respondents can run it before spending anything.

(3) A twin-to-twin linkage channel within a shared pipeline — governed by which element differs, not how many — and a cross-pipeline test our own instrument could not carry. Two twins of the same person, sharing their persona source and differing only in model or output format, designate each other when they share enough items — while the attacker holds no real human answer of any kind. On Twin-2K-500: 36.4 % mean top-1 over 30 pairs sharing 60 common items, against an anti-artifact control of 0.04 %. On the Park archive: 11.9 % [10.1 ; 13.8] and 13.0 % [11.2 ; 14.9] in the two directions at 177 common items, against a control of at most 0.31 %. We never report a single number for this channel without its common-item count. At one pipeline element changed, the rate runs from 17.2 % (model) to 81.4 % (decoding), and the *distance law* we preregistered — leakage falling with the number of differing components — is refuted by our own measurements (§ 5.4). Anonymeter's contrary conclusion [18] is reached in a regime it did not test (§ 2.1). Built end-to-end independently instead (T2, § 3), top-1 falls to 1.8 % [0.4 ; 3.6] — but that arm is **not interpretable**, its own twins identifying the real person only at chance, so T2 remains untested.

(4) The leakage is localised in the pattern of answers, not in memorised content. An ablation destroys the signal by permuting the order of a twin's answers while leaving their content untouched — 33.1 % to 0.046 % top-1 (§ 5.5): what identifies is the dependence structure between answers, not any answer's content, which is what a regurgitation account would require. The pipeline also trains nothing on the target population, so no train/test gap exists to exploit, unlike Yeom et al. [41] and Feldman [14]. We state the limit plainly: our training-cutoff control lives in a regime where none of our twins identifies almost anyone (§ 5.7), so **we do not claim a measured dissociation between memorization and re-identification.** The narrower claim we do make — a structural mechanism, localised by ablation — is what separates our channel from the memorization [7] and free-text inference literatures [23, 24, 35].

(5) Open-world rates with their human ceiling, and an attack strong enough to falsify our own first measurement. Closed-world 1-in-N rates assume the target is in the pool; the defensible measurement drops that assumption and reports true detections against false accusations (§ 5.3, Figure 1). A rarity-weighted likelihood attack with out-of-fold parameters raises the Park archive's closed-world top-1 from 65.51 % to 90.40 % and triples its open-world rate at 1 % false accusations — a correction of our own published figure (§ 5.3, § 5.8).

(6) A defense measured with its real cost, and held against an adaptive attacker. Not a new mechanism — a variant of the 1998 Post Randomisation Method [20] — but applied to LLM twins with a risk-utility curve that is measured rather than assumed, reported with the cost component a single average would hide, and tested against an attacker who knows how the defense works (§ 6). Against our own preregistered prediction, differential privacy is *not* dominated by it on aggregate utility (§ 6.2).

(7) A bits-of-identity instrument that transports across datasets where the raw rate does not. A derived instrument, not a discovery: a case of Rényi min-entropy leakage [34], in the spirit of Eckersley's surprisal [13], addressing the transportability problem of the Rocher et al. scaling law [31] by another route (§ 5.6).

1.4 Negative results, stated in front

Fourteen preregistered predictions were refuted, two are inconclusive, and one could not be tested at all (Table 1, § 7.1); three later verdicts — the distance law (§ 5.4) and the third dataset's two predictions (§ 5.8) — sit outside that table for the reasons given there. Our own twins identify almost nobody, and the constructive demonstration meant to close that gap — raise fidelity with a frontier model and watch the leakage return — **failed**: `openai/gpt-4.1` on Twin's per-item recipe left fidelity at 0.1714 against their 0.708 (§ 5.7). We still do not know what the Twin-2K-500 recipe does differently. And the coupling that motivated this work does not survive its own control (§ 5.1). We consider a reader who stops here to have read the paper honestly.

2 Background and Related Work

2.1 Privacy of synthetic data

Stadler, Oprisanu and Troncoso show that generated tabular data remains linkable to its sources [36]. Giomi et al. formalise three attacks — singling out, linkability, inference — and conclude, on their datasets, that linkability is the weakest risk [18]. That conclusion is contested: Annamalai, Gadotti and Rocher [2] and Ganev and De Cristofaro [15–17] show reconstruction attacks defeating distance-to-closest-record metrics; Yao et al. [40] and Meeus et al. [25] generalise the critique; Golob, Pentyala and De Cock make these attacks an emerging standard [19].

What distinguishes us. This corpus concerns tabular microdata from classical generative models, evaluated by continuous similarity metrics. Our object is a vector of categorical answers produced by an LLM conditioned on a person, where no similarity metric substitutes for a direct matching attack.

2.2 Re-identification of real data

Narayanan and Shmatikov re-identify Netflix users [26]; de Montjoye et al. show four spatio-temporal points suffice on credit-card metadata [9]; Rocher, Hendrickx and de Montjoye model risk on incomplete samples [30] and extend it into a scaling law [31]. Taub et al. ground the statistical measurement of disclosure risk on synthetic data (CAP, TCAP) on the premise that synthetic generation breaks the link between identity and datum [37].

What distinguishes us. These works re-identify from genuine auxiliary data. Our attack's input is never a real datum of the target: it is a model output generated from a persona, compared against real answers the attacker separately holds. We falsify precisely the CAP/TCAP premise, on LLM-simulated survey microdata.

2.3 Privacy and LLMs

Carlini et al. document memorization and regurgitation of training sequences [7]. Staab et al. infer personal attributes from innocuous free text [35]. Ko et al. de-anonymise online authors by multi-cue agentic inference [23] — the closest threatening neighbour — and Lermen et al. demonstrate the same attack at scale [24]. Theoretically, Yeom et al. connect membership attacks to overfitting [41]; Feldman shows memorizing rare cases is necessary for generalization [14]; tracing codes [5, 12] are the closest formal parent to our 1-in-N attack, bounding the worst case of an optimal adversary recovering an individual from aggregate statistics, though in dimension far exceeding the number of individuals.

What distinguishes us. Those works all start from free text bearing direct semantic cues, aggregated by an agent reasoning over it. Our channel is narrower and drier: closed-choice categorical answers, no free text, no agentic reasoning, and a mechanism localised in the dependence structure rather than in memorised content (§ 5.5). Unlike Yeom and Feldman, our pipeline trains no model on the target population.

2.4 Respondent simulation and twins

Argyle et al. launched the use of LLMs as panel substitutes [3]; their Dataverse archive supplies the 2023 end of our trajectory, and our measurement does not contradict their paper, whose study 3 compares matrices of Cramér's V — the association structure of a sample, not individual accuracy. Park et al. cite privacy in restricting access to their 1,052 agents' individual responses [27], while the publicly posted replication package nonetheless carries individual participant and agent responses (Ethical Considerations) — a risk anticipated, never measured. Toubia et al. publish the 2,058-twin panel we attack, treating neither privacy nor linkage [38]. Peng et al. document group-fidelity distortions on that same panel without connection to re-identification [28]. Bonagiri et al. issue a normative call to evaluate these risks without measuring them [4]. None of these works measures a re-identification rate from twin outputs. That is the gap we fill.

2.5 Closest prior art and contradictors

Ward et al. test 13 attacks over 9 generators and 48 tabular datasets, and state in prose a relation between quality and leakage close to ours [39]; Byun et al. observe a near-monotone frontier of the same type without a coefficient [6]. Three differences separate them from our claim: their quality is *aggregate* (MMD, Jensen-Shannon divergence, downstream AUC), never a correspondence to the right person; their risk is *membership* in a training set, not 1-in-N identification; and they publish no coefficient.

Three contradictors hold that fidelity and risk separate: Platzer and Reutterer [29]; Adams et al., for whom a generator without formal guarantees preserves fidelity and utility without evident privacy harm [1]; and Shafieinejad et al., observing decoupling as quality saturates within a single tabular diffusion model trained longer [33]. Our response is not that they are wrong: all three measure *aggregate* fidelity, never individual correspondence to the right person, and none tests whether the coupling they observe survives a null matched on per-person accuracy. Ours does not (§ 5.1). Their decoupling also unfolds over training time within one model, where ours is a cross-sectional correlation across twelve heterogeneous methods against 1-in-N re-identification.

2.6 Defenses

Our D4 is not new. Shuffling answers between units of the same segment is a variant of the Post Randomisation Method [20], which perturbs categorical variables through a known transition matrix leaving margins invariant in expectation, and of data swapping — the same family as the partially synthetic data tradition of Reiter and Drechsler, extended by Drechsler [10] and Bowen et al. [22]. What is new is the application to LLM twins with a measured risk-utility curve (§ 6).

3 Threat Models

We distinguish three scenarios, assuming the first one's strongest objection rather than deflecting it.

3.1 T1 — Panel holder publishes unkeyed twins; attacker holds the real answers

A panel holder releases per-person twin records without the key linking each twin to its respondent. An attacker who holds **the real answers to the same items** — an insider, a panel co-owner, a leak of the source file — matches twins to respondents. This is a *linkability* result in the sense of the Article 29 Working Party and the GDPR, not identification from public information: the quasi-identifier required is a specific questionnaire, rarely held outside the panel.

We state plainly what this is not. It does not show that a third party recovers a named person from public information. And on Twin-2K-500 itself the matching is trivial without any attack — `pid` equals `TWIN_ID`, and the twin files copy wave metadata line by line from the humans. The result therefore documents a generic risk for anyone releasing "anonymous" twins without that key, a practice Twin-2K-500 does not follow. The objection "this is not real re-identification, the attacker already holds the true answers" is assumed, not rebutted; our answers are the explicit model above and the open-world measurement of § 5.3. A third, T2, would remove the assumption entirely — but we did not manage to test it.

3.2 T2 — Two organisations publish twins of the same cohort; attacker holds nothing real

If several organisations each publish their own twins of the same panel with sufficient item overlap, a third party cross-references them holding no human answer at all. This threat model has no direct precedent in the synthetic-data literature: Anonymeter's linkability presumes real attribute values in hand [18], and Guépin et al. remove the real-auxiliary-data assumption but target membership in a single generator's training set, never matching between two independent generations [21].

We tried to test it directly, and could not. § 5.4's 36.4 % twin-to-twin top-1 is a same-team, same-persona-file measurement and does not speak to T2 as stated. Built as T2 actually describes, two pipelines sharing nothing but the target person give top-1 1.8 % [0.4 ; 3.6] — but both fail the interpretability control of § 4.5, so the contrast opposes two noise sources. **T2 is neither supported nor refuted: it was not tested** (§ 5.4).

3.3 T3 — Pre-publication self-audit

A panel holder runs the attack against their own file before publication, to decide whether a defense is required — the constructive use of the method, and the one the artifact (Open Science) serves.

3.4 Attacker capabilities, all models

The attacker performs no training on the target population, has no query access to the generating model, and uses only rank-based matching over categorical vectors.

4 Data and Methods

4.1 Datasets

Twin-2K-500 [38], 2,058 respondents, public under CC BY 4.0. Target: wave 4, 60 items usable in common out of 108 (40 purchase items of the form "would you buy this product at this price?", 20 opinion items). Attacker context where applicable: 494 context items from waves 1–3 plus 14 demographic variables. Chance top-1 in the closed pool is 0.049 %. A human test-retest condition gives the empirical ceiling: 81.6 % [79.9 ; 83.2] top-1, accuracy 0.745.

Park et al. archive [27], 1,052 participants, 177 common items, replication package publicly posted. Of its five agent conditions we analyse four — composite, interview-only, survey-only, demographic — the `persona` condition being excluded by the interpretability control (§ 4.5). Human retest ceiling 96.8 %.

Argyle et al. archive [3], Harvard Dataverse, **CC0 1.0**: 4,270 ANES 2016 respondents, of whom **2,148** answer all **12** matched items; chance top-1 is 0.047 %. Each twin column is a GPT-3 (`davinci`) prediction produced while holding that item out and supplying the person's eleven other **true** answers, so these twins are *better* informed than our persona twins, not worse: that confound runs against us. Three twin versions of the same people exist (main decoding, temperature 0.01 and 1.0); they vary decoding, not epoch, and we never present them as survey waves.

No new data was collected. The attack uses only already-published content.

4.2 Methods compared

Twelve heterogeneous methods carry the quality-leakage analysis: 8 LLM twins and 4 statistical reference points (Demographics Only, B1 argmax over 14 demographics, B2 argmax, PMM k=10). Additional comparators appear in § 5.2: logistic regression on context, a k=1 donor on context, two individualised *generators* (sampled LR, Gaussian copula), and an explicitly advantaged k=1 donor fitted directly on the target block. Sequential CART and a 300-tree forest were tried before that donor and abandoned at 0.45–0.53 accuracy: the retained comparator was not chosen to lose. On the Argyle archive the comparators are **B-demo**, a leave-one-out imputer on the four demographic items, and **B-oracle**, the same imputer given the person's eleven other true answers — the comparator matched to that archive's own conditioning.

4.3 Metrics

Attacks. Two matching rules are used throughout, and every rate is labelled with the one that produced it. The **naive** attack ranks candidates by Hamming-type agreement over common items. The **strong** attack (A-LLR) ranks them by a rarity-weighted log-likelihood whose parameters are estimated **out-of-fold over 5 folds**. A third variant (A-MI) scores higher still on the Park archive (93.25 %) but estimates its weights **in sample**, an advantage declared in the preregistration, so we never report it as our result. An **adaptive** attacker, used only against the defense (§ 6.1), additionally knows the mechanism and which items it leaves untouched.

Closed-world top-1 and top-10: rank of the true target among all N candidates by the rule stated. *Open-world*: true-positive rate against false-positive rate, where the system may decline to accuse (§ 5.3). *Bits of identity*: log2 of the ratio of correct one-shot guessing probabilities before and after conditioning on the persona — a Rényi min-entropy leakage quantity [34]. *Imitation quality*: the share of the human test-retest floor attained.

All intervals are 95 % bootstrap confidence intervals unless stated.

4.4 Preregistration and self-refutation

Attack, defense, mechanism, the two archive replications, the disjoint-item test and the distance law were each preregistered before computation, and we report every refuted prediction in Table 1. A census found **no inversion** between a plan and its result across the 28 verifiable pairs; for the large majority the ordering is provable from version-control history, while a small number were committed alongside their results or left unversioned, which we state as a limit rather than claim an unbroken chain. The plans are not yet third-party timestamped.

4.5 Validity controls

Provenance. On Twin-2K-500, the context contains 0 wave-4 columns and 0 wave-4 QIDs; maximum normalised mutual information between context and item is 0.16 against 0.31 for the retest on the same item. On cells where wave 4 differs from waves 1–3, the JSON 4.1 output equals wave 4 in 37.7 % of cases and waves 1–3 in 37.9 % — symmetric, so no indication of copying. On the Park archive, over 34,915 cells where the human changed their mind, the composite follows wave 1 in 41.6 % and wave 2 in 43.2 %, and the 4.3–4.9 point wave-1 bias of the other three conditions holds for the `demographic` condition too, so it is a shared calibration artifact, not item leakage. On the Argyle archive each twin column is generated with its own item held out, so no tautology exists on either side; the same holds for B-demo and B-oracle, whose target item is removed from their context.

Empty-run validation. Uniform ranks yield 0.00 bits; the B0 mode baseline yields −0.004 [−0.003 ; 0.002] bits.

Interpretability control, adopted after it cost us an experiment. We state as a rule the control whose absence invalidated two of our own arms (§ 5.4): **no experimental arm is interpretable until its candidate pipeline has been shown to beat the demographic baseline in top-1 against the real human answers**, on the pool and items actually attacked, with the baseline recomputed on those same rows rather than carried over as a constant. The check consumes no model call and no budget, and it must precede the first paid contrast; a pipeline that fails it yields twins whose contrasts oppose noise to noise and return chance whatever is manipulated. Applied since, it has excluded our own pipeline B, all three GPT-3 twins of the Argyle archive (§ 5.8), and the Park archive's `persona` condition, whose agent — built on the person's own self-description — re-identifies her at 0.26 % [0.06 ; 0.52] against a demographic baseline of 0.39 %. Baselines are always taken at the pool size of the arm they judge, being strongly pool-dependent: 2.13 % at 2,058, 9.20 % at 200, 13.29 % at 120.

5 Results

5.1 The coupling, and the null that absorbs it

Across 12 heterogeneous methods, the ordering by imitation quality and the ordering by re-identification rate nearly coincide: Spearman 0.958 [0.930 ; 0.993], n = 12, robust to restriction to LLMs over a narrow fidelity range (rho = 0.976) and to jackknife (0.958–0.986).

The preregistered disjoint-item test measures fidelity on one half of the 60 items and leakage on the other, over 50 stratified random A/B splits: Spearman **0.969**, 95 % CI **[0.937 ; 0.993]**, with 100 % of splits above 0.7. Prediction (a) confirmed — the relation is not an artifact of recycled computation.

Prediction (b) was refuted, and the witness that refutes it has since been rebuilt to remove a real defect without changing the verdict. The original marginal null did carry that defect: its *wrong* cells avoided the person's true answer (`c7_disjoint.py` l. 133), inflating its leakage by a factor 1.3 (41.2 % against 31.6 % once wrong values are drawn from the population marginal). With the defect corrected, the witness still reproduces the coupling — **rho 0.974, 5th–95th percentiles [0.950 ; 0.993], against 0.965 observed** over the 12 configurations, measured on whole items (the 0.969 above being the disjoint-split value of the same quantity; Figure 2) — so **prediction (b) remains refuted**.

This witness is not an information-free object, and we do not describe it as one: at matched accuracy it leaks **more** than our twin (31.6 % top-1 against 20.7 %). Five further witnesses, matching that accuracy against a *surrogate* target rather than the person, fall to chance on both axes; the observed rho exceeds them, but that margin bears on nothing, their null distribution being that of Spearman under total absence of information at n = 12.

We therefore do not claim, anywhere in this paper, that a twin's fidelity to its person causes its leakage, nor that fidelity and identifiability are one axis, nor that at equal accuracy every twin leaks more than every predictor: the residuals of the regression on accuracy do **not** separate the families, 3 of 7 LLM configurations falling below the line. One positive point remains testable outside this circularity: the human point, in raw position, dominates on both axes — fidelity 0.303, leakage 0.520 — against 0.154 and 0.067 for the best twin, so there is no saturation at the top of the range.

Limits. Twelve points. Raw accuracy is only a noisy proxy (r = 0.72, p = 0.008) for the right axis, and Spearman is coarse at n = 12 — the null touches 1.000 through small-sample effects — so the verdict concerns the order of magnitude, not the third decimal.

5.2 Closed-world rates and the comparator question

At comparable marginal accuracy, no non-LLM predictor we tested exceeds 0.3 % top-1 where the twin reaches 20.7 % [19.0 ; 22.4] under the naive attack, and **23.23 % [21.5 ; 25.0]** under the strong one — **in a closed world, with the target always present in a pool of 2,058**. The comparator figures below are naive-attack rates, which makes the contrast conservative. JSON Persona GPT4.1 attains that rate at accuracy 0.590; the comparators, at accuracy 0.457–0.511: PMM k=10, 0.23 % [0.06 ; 0.42]; B2 argmax, 0.07 % [0 ; 0.19]; k=1 donor on context alone, 0.13 %; LR on context, 0.22 %. Two individualised *generators* — sampled LR and Gaussian copula — give 0.25 % [0.11 ; 0.41] and 0.22 % [0.11 ; 0.36], answering the objection that we compare an individualised generator against predictors that regress to the mean.

We then gave a comparator an explicit advantage: a k=1 nearest-neighbour donor fitted **directly on the target block's answers** — seeing what the LLM twin never sees. It reaches accuracy 0.584 against the twin's 0.590, carries genuine individual signal (0.584 true against 0.442 under within-segment permutation), and despite that advantage obtains a top-1 of **0.00 % [0 ; 0.18]** (Clopper-Pearson, n = 2,058).

This claim does not generalise to every leakage metric, and top-10 runs the other way. That same advantaged comparator reaches a top-10 of **53.2 % [51.0 ; 55.3]**, *above* the twin's 42.7 %: the direction of the whole contrast depends on the choice of k, which no principle fixes at 1. We claim neither that all individualised prediction identifies nor strict accuracy equality — the generators plateau at 0.476. Neither synthpop/CART nor CTGAN was tested.

5.3 Open world — the defensible measurement

Removing the closed-world assumption gives the number we consider hardest to attack in review. At a false-accusation rate of 1 %, the strong attack (A-LLR, parameters estimated out-of-fold over 5 folds) recovers the right person **60.17 % [54.5 ; 64.4]** of the time on the Park archive and **4.28 % [3.3 ; 5.6]** on Twin-2K-500, under a person-level bootstrap that **re-estimates** its parameters in each resample while keeping every copy of a person in the same fold. At FPR = 0.1 % these fall to **44.37 % [23.3 ; 53.8]** (Park) and **1.01 % [0.05 ; 2.82]** (Twin) — intervals this wide because the threshold there rests on roughly **one** absolute false positive on Park and two on Twin, so we report them as very unstable estimates, not as measured rates. Statistical comparators stay indistinguishable from noise at both FPRs. The human ceiling at FPR = 1 % is 54.5 % [50.8 ; 57.5] and 90.7 % [88.2 ; 93.3].

These figures replace the ones we first published, and the correction runs in both directions. The naive attack gave 3.04 % [2.0 ; 4.0] and 20.39 % [15.7 ; 24.5] at 1 % FPR. On the Park archive the strengthening is statistically clear — no overlap at either FPR, and the rate **triples**. **On Twin-2K-500 the apparent gain does not survive its interval**: 3.04 → 4.28 % overlaps broadly, so we do not present the strong attacker as identifying better than the naive one here. Park's closed-world gain is a different quantity, untouched by this comparison.

Our preregistered prediction was refuted, and one half of it was then reversed by a better attack. We announced > 5 % (Twin) and > 30 % (Park); under the naive attack both fell below the bar. Under the strong attack Twin (4.28 %) still falls below, while Park (60.17 %) passes the bar it had failed.

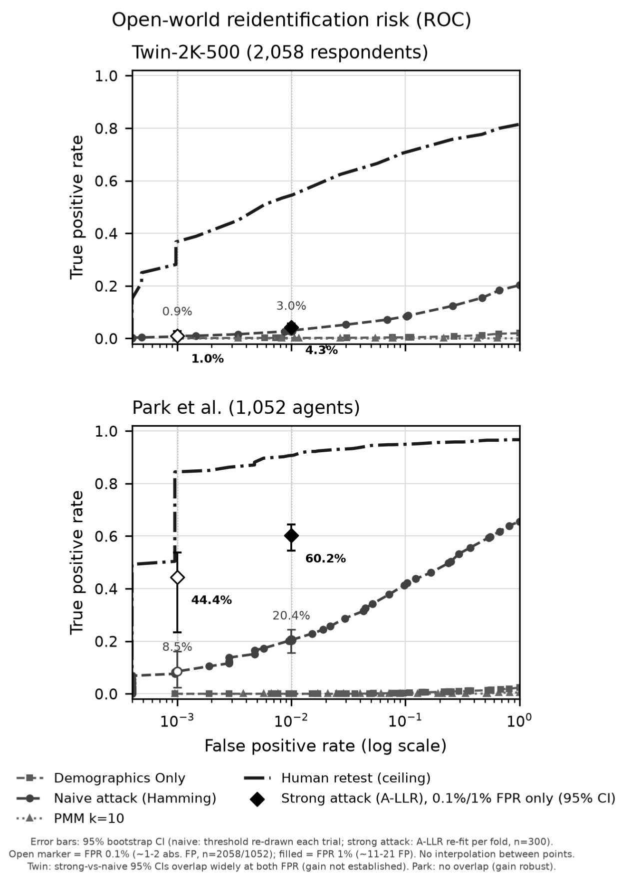

Figure 1: Open-world detection, naive and strong attack. Two panels, one per dataset, false-accusation rate on a log scale against true-detection rate. Each overlays the naive Hamming attack as a full ROC curve with the strong A-LLR attack as isolated diamonds, measured at only two thresholds (FPR = 0.1 % and 1 %): no curve should be read into the diamonds. Statistical comparators sit near zero, the human retest ceiling above. *At a glance*: at a tenable false-accusation rate the risk is real and far above the comparators under both attacks, and the strong attack lifts the Park archive close to the human ceiling — with no causal reading of the quality-leakage coupling required. Data: `resultats/c7-monde-ouvert-roc.csv`, `resultats/c7-attaquant-fort.csv`.

5.4 The twin-to-twin channel within one pipeline, what governs it, and why our T2 test does not count

Two twins of the same person designate each other when they share enough items, with no real data on the attacker's side. The measurements below share one team and, for Twin-2K-500, one persona source per person: they gauge the channel within a pipeline, not the cross-organisation threat model T2 (§ 3).

Twin-2K-500, tabulated by common-item count. 30 pairs sharing 60 common items: mean top-1 **36.4 %**, maximum 83.58 %. 12 pairs sharing 19 common items: mean top-1 **0.45 %** (chance 0.05–0.10 %). Demographics Only against the rich twins gives 7.76 %. The anti-artifact control — a decoy from the same segment — gives 0.04 % on the 60-item pairs (factor 900) and 0.10 % on the 19-item pairs.

Park archive replication: 1,052 agents, 177 common items. Interview→survey top-1 **11.9 % [10.1 ; 13.8]**; survey→interview **13.0 % [11.2 ; 14.9]**; top-10 45–46 %. Demographic bound 1.5–2.5 %, anti-artifact decoy control **at most 0.31 %**. **Preregistered prediction refuted**: we set a threshold of at least 20 % top-1 there, and 11.9–13.0 % does not reach it — though the preregistered *failure* criterion is not met either, at a factor of 5–8× over the demographic bound.

We never report a single figure for this channel without its common-item count; a previously circulated average of 26.11 % pooled the two item regimes and is withdrawn. Nor do we claim leakage crosses waves: that arm is infeasible, with 0 items in common.

The channel is not an artifact of a shared prompt template or of demographics. Decomposing agreement between two twins of the same person over the 60 common items: the floor between *strangers* is **46.0 %**, the demographic-ideological segment adds **3.8 points**, and the person adds **17.4 points** — 82 % of the rise above the floor. A pure-population witness identifies the right person **0.07 %** of the time against **36.4 %** for the real attack.

The cross-dataset anomaly, explained in part. The Park archive shares 177 common items — roughly three times the 60 of Twin-2K-500 — and leaks roughly three times *less*. Item count is the wrong denominator: the 177 GSS items are strongly redundant (mean Cramér's V 0.1055 against 0.0834), giving ≈ **9.0 effectively independent items against** ≈ **10.1** on Twin — a real ratio of **0.89, not 2.95**. Pool size runs the wrong way too: shrinking Twin's pool to 1,052 *raises* its top-1 (37.3 → 44.1 %), while Park already has the smaller pool. **A residue of roughly 3× remains unexplained** at comparable effective information, and we declare it: the remaining candidates — persona format and length, and the different generating pipelines — are not testable on the material we hold. The same denominator places the third dataset far below both, at **4.31** effective items for 12 raw ones (mean Cramér's V 0.162), 43 % of Twin's effective information (§ 5.8).

Limits. The 12 low pairs are exactly those at 19 common items: the effect follows the shared-item count, not an aberrant configuration. On Park the item-rate relation does not follow Twin's.

T2 itself: two genuinely independent pipelines, and a test that does not count. Two pipelines sharing nothing but the target person — `deepseek-v4-flash` structured JSON dossiers versus `qwen-2.5-72b` narrative biographies, different templates and output formats — matched head to head on the same 60 items (n = 142 of 200 planned, stopped by an OpenRouter transport error; an earlier stop at n = 82 was reconciled and confirmed unbilled). Preregistered rule: T2 survives if the top-1 CI excludes the segment control *and* stays at least 2× the demographic baseline. Result: top-1 1.8 % [0.4 ; 3.6], against a demographic baseline of 9.2 % computed on the enclosing pool of 200 — the baseline matched to this arm's own pool of 142 was never computed and lies above that.

We no longer read this as a refutation of T2. The witness is built on the pipeline of the paragraph below and shares its defect: neither arm produces twins that identify the real person better than chance (0.0 % and 0.8 % top-1 against 0.83 % expected on a pool of 120, where the Twin team's own twins reach 20.2–38.9 % on that same pool). The comparison therefore opposes two noise sources, and would have returned chance whether or not a cross-organisation channel exists: it measures our instrument, not the world. The 36.4 % figure above, being a same-pipeline measurement, is not evidence for T2 either, so the threat model stands open in both directions.

Which element differs, not how many — and a preregistered law our own measurements refute. Among the six Twin-2K-500 configurations that pass the interpretability control (2,058 people, 60 common items, symmetric attack), changing a single published pipeline descriptor gives **81.4 %** [80.0 ; 82.8] for decoding alone, **67.1 %** [65.3 ; 68.7] for the prompt template alone, **33.5 %** [31.6 ; 35.3] for reasoning alone and **17.2 %** [15.8 ; 18.6] for the model alone. We preregistered a *distance law* — leakage falling with the number of published descriptors that differ — and our own measurements refute it: Spearman −0**.232** [−0.321 ; −0.187] over 15 pairs, and **+0.003** [−0.047 ; +0.068] once the individual fidelity of the two compared twins is controlled, against the −0.7 predicted. At a distance of one, rates span 17 % to 81 %: the spread *within* a distance level dwarfs the spread *between* levels. Counting changed components predicts nothing useful; naming them does. On the Park archive the same control leaves only three pairs, so no replication is possible. Two limits bound this: the distance weights are arbitrary, and we would not rescue them by refitting on the linkage rate, which would make the measure circular; and the independent unit is the configuration, not the pair — 18 pairs, 9 configurations, two teams.

One-factor-at-a-time from our own pipeline: arithmetically sound, not interpretable. Varying model, template and persona format one at a time from our baseline pipeline B gives 0.67 %, 0.83 % and 2.65 % (n = 120, chance 0.83 %). **The interpretation is not available to us**: pipeline B and all three derived conditions identify the real person at chance, so each contrast opposes two noise sources. We withdraw the reading we first gave these numbers — that changing any single component collapses the channel — and with it the refutation of the matching preregistered prediction: an arm that could test nothing refutes nothing. We also correct a comparison of our own: these twin-to-twin rates were set against 13.29 %, a *twin-to-human* baseline, where the homogeneous twin-to-twin baseline on that pool is 23.69–34.42 %.

5.5 Mechanism: being right while deviating — and three refuted hypotheses

What identifies is neither the frequency of departures from the modal answer nor answer diversity, but their **correctness** — and the whole response vector is required. Correctness of deviations: 68.5 % for the human retest (leakage 81.6 %), 50.3 % for JSON 4.1 (20.6 %), 43.3 % for Demographics Only (2.14 %), 39.1 % for PMM

(0.22 %); PMM deviates and diversifies as much as the twin without leaking.

The ablation is the strongest evidence: permuting the order of each twin's 40 purchase answers drops closed-world top-1 from **33.1 % [31.2 ; 35.2] to 0.046 % [0 ; 0.11]**, below chance, and an oracle given only the count of "yes" answers gives 0.29 %. The identifying quantity is the dependence structure between answers, carried by the conditioning — not any answer taken in isolation.

That ablation starts from a higher rate than this paper's headline 20.7 % because it attacks the **40 purchase items alone**, where the headline rate uses all **60 common items**: adding the 20 opinion items *lowers* top-1, those items identifying 46× less and diluting a naive Hamming rule rather than helping it. That is a change of attack surface, not a discrepancy.

Three preregistered hypotheses refuted. H1 (entropy): opinion items carry 2.10 bits against 0.99 for purchase items and identify 46× less — top-1 0.24 % [0.07 ; 0.46] against 11.0 % [9.9 ; 12.3] for the purchase subset matched to them on entropy, a 20-item comparison. H4 (stereotypy): 36.2 % [35.6 ; 36.8] against 37.5 % [36.9 ; 38.1] in humans. And "deviations alone carry $\geq$ 80 % of the leakage": 0.68 % in the mixed condition. We do not present "wrong deviations alone = 0.0 %" as a result: it is a floor of the method.

Limits. The mechanism is established on Twin's purchase block alone, and is in apparent tension with the Park archive, where opinion items identify most. Two simple explanations are excluded: at 20 items the Park archive gives 11.7 % against 0.24 % for Twin's 20 opinion items, **at lower per-item entropy** (1.30 against 2.10). What remains is consistent with the block effect, which we present as the **best-supported hypothesis, not a result**.

5.6 Bits of identity: the instrument transports, the rate does not

An LLM twin costs approximately 0.4 bits of identity per point of accuracy gained, against 0.068 for the best useful statistical comparator. Twin JSON 4.1: **3.55 bits [3.39 ; 3.72]** against a ceiling of $\log_2 N = 11.01$. Park composite: **7.47 [7.27 ; 7.66]** against a ceiling of 10.04. PMM: 0.19 [0.16 ; 0.23]. B2: 0.08.

Transportability is the point. Bits normalised by human entropy give 0.0439 (Twin) against 0.0327 (Park) — a factor of **1.34** — where top-1 varies by a factor of **3.2**. Demographics Only and PMM leak with **no accuracy gain at all** (−0.5 and −2.4 points), a case that a ratio alone would mask. The instrument's range of validity is bounded on the other side by the third dataset, where a twin that carries no demonstrable individual information has no ratio worth transporting (§ 5.8).

A nuance against our own hypothesis. The human retest has a ratio of 0.37 (Twin) and 0.44 (Park) — as good as or better than the twins. The excess cost exists relative to statistical predictors, **not relative to a human answering twice**.

Limits. The bits are a **lower bound** (dyadic bound plus Miller-Madow). The sum of per-item mutual information is unusable as a total: 58.7 bits against a ceiling of 10.04 on the Park archive, as preregistered.

5.7 The risk depends on the recipe — and we could not reproduce it

Our own twins, produced in a single call to cheap models from a raw profile, identify almost nobody: 0.00 % to **0.83 % [0 ; 2.15]** (deepseek-v4/R1) across 7 cells, against 2.13 % for the other team's Demographics Only. That interval **contains** 2.13 %, so no precise factor is established — only the order of magnitude, that no model reaches 1 %. Any previously circulated "25× below" figure is withdrawn. Memorization is not the explanation here: llama31-8b, with a training cutoff preceding publication, copies nothing verbatim and leaks no more than the models that do copy — a control on our own twins, in a regime where none of them identifies almost anyone, which we do not extend to the published twins of § 5.2–§ 5.4 (§ 1.3).

Two further preregistered predictions refuted. (i) R1 $\geq$ 10 % on at least 2 of 3 models — no model reaches 1 %. (ii) Call granularity would explain the leakage — not confirmed (underpowered, especially at n = 10 for the per-item arm): single call 0.00 % [0 ; 8.8] and per-item call 0.00 % [0 ; 30.85], with top-10 running in the **opposite** direction to the prediction (7.5 % against 0 %). We therefore do not claim that leakage comes from the call format, nor that the cause is identified. **It is not.**

We then paid to test the obvious explanation, and it failed. Raising fidelity with a frontier model to see whether the leakage returns was run on `openai/gpt-4.1` (Twin's own model class) using Twin's per-item call recipe without modification: 30 persons × 60 items = 1,800 calls, 100 % parse rate, 4.9949 USD of actual cost. Accuracy reached **0.4722 [0.4361 ; 0.5050]** against Twin's 0.574; fidelity **0.1714** against their 0.708; top-1 and top-10 **0.00 % [0 ; 11.57]** (Clopper-Pearson, n = 30). **One further preregistered prediction refuted**: accuracy > 0.55; top-1 > 5 % is **not refuted**, the exact interval containing the 5 % threshold — inconclusive for lack of power. Fidelity > 0.10 was confirmed at 0.1714, barely above our best cheap twin (0.177): the frontier model bought us almost nothing. We did not make the leakage come back, because we did not reach Twin's fidelity — and it is that fidelity gap, not the underpowered top-1, that carries the conclusion. At the fidelity actually reached, the regression over the 12 points of `c7-compromis.csv` (slope 0.1834, r = 0.785) predicts leakage of **1.10 %** [−9.15 ; 11.35] and the observed 0.00 % falls inside it: coming from a new model *and* a new recipe, the point follows the curve rather than breaking it.

The unknown narrows instead of disappearing. Neither the model nor the call granularity, what remains implicated is the **format and length of the profile**: Twin's full JSON persona runs to roughly 121,000 characters against our 8,000-character truncation.

Limits. n = 40 persons for the recipe arm, truncated to n = 10 on the per-item arm, and n = 30 on the paid frontier-model arm; a floor effect is not excluded. Roughly 1,490 calls first failed with HTTP 402 from in-flight credit reservation, not lack of balance, and were retried at decreasing concurrency without double billing.

5.8 A second dataset, and a third at the other end of the trajectory

On the Park archive, the composite agent designates the right participant among 1,052 **in a closed world** in **90.40 % [88.6 ; 92.2]** of cases under the strong attack, top-10 97.8 %, against a human retest ceiling of 96.8 %. **This figure replaces the 65.51 % [62.7 ; 68.3] we first published**: a naive agreement attack understated this archive's leakage by 38 % in relative terms. An in-sample information-weighted variant reaches 93.25 %, which we do not claim as our result (§ 4.3).

The remaining conditions are **naive-attack measurements**, the strong attack not having been re-run on them: interview-only **44.7 % [41.8 ; 47.5]**, survey-only 20.6 % [18.2 ; 23.0], demographic 2.26 %. The interview-only figure carries zero documentary exposure to the survey, so it is the cleanest demonstration that an agent built from a conversation alone identifies its person — but given what the strong attack did to the composite, **44.7 % must be read as a lower bound**.

We do not compare the Park and Twin rates as measurements of the same thing: item count (177 against 60), per-item entropy (1.30 against 0.99) and human ceilings (96.8 against 81.6) all differ. Equalised at k = 60, Park gives 34.0 % [23.5 ; 51.5] against Twin's 20.7 % — naive attack, 20 draws, a very wide interval.

The third dataset, and the analysis that stops before it starts. On the Argyle archive the interpretability control **fails for all three GPT-3 twins**, and the preregistered decision rule halts the analysis there: top-1 0.14 % [0.05 ; 0.27], 0.11 % [0.04 ; 0.20] and 0.09 % [0.02 ; 0.22] against a demographic baseline of **0.10 %** [0.02 ; 0.23] recomputed on that same pool of 2,148, all intervals overlapping. No contrast is interpreted, and the anchor-pool draws were never computed: they would have interpreted noise. The verdict rests on no choice of items — dropping age gives 0.08–0.13 %, the eight attitude items alone 0.09–0.11 %. The reason is visible per item: mean exact accuracy **47.2 %**, below the demographic imputer's 51.5 % and below the modal answer on nine items of twelve. In bits, the best of the three carries **0.094** [0.074 ; 0.118] of an 11.07 ceiling — the level of Twin's *weakest* statistical comparator and nine times below Twin's demographics-only twin — while a nearest neighbour given exactly the same eleven true answers carries 0.342 [0.301 ; 0.389], **3.6 times more**.

Two preregistered predictions were missed, and we count them as neither. We had announced top-1 2.5 % [1.0 ; 6.0] and normalised bits [0.020 ; 0.070], derived from our measured item and pool curves; the observations, 0.14 % and 0.0044, fall far outside both. Neither our scaling law nor the transportability claim is refuted by this, because neither is testable here: both presuppose a twin that carries individual information. **We cannot distinguish "our law overstates leakage at a low effective item count" from "this twin is at the noise level", and we do not pretend to.** Both rows therefore sit outside Table 1's count, like the distance law of § 5.4 — three verdicts reached after the census of § 7.1 was closed.

5.9 Scale

From N = 50 to N = 2,058, the twin/demographic ratio grows from 2.8 to 9.7: top-1 falls from 53.2 % [50.0 ; 56.3] to 20.7 % [20.7 ; 20.8], and the share of the human ceiling from 57.0 % to 25.4 % — naive-attack rates, the scale study not having been re-run under the strong attack. **We extrapolate no value beyond N ≈ 4,000**: a power law and a logarithmic law fitted on the same 6 points already diverge at twice the range (18.1 % against 13.9 %).

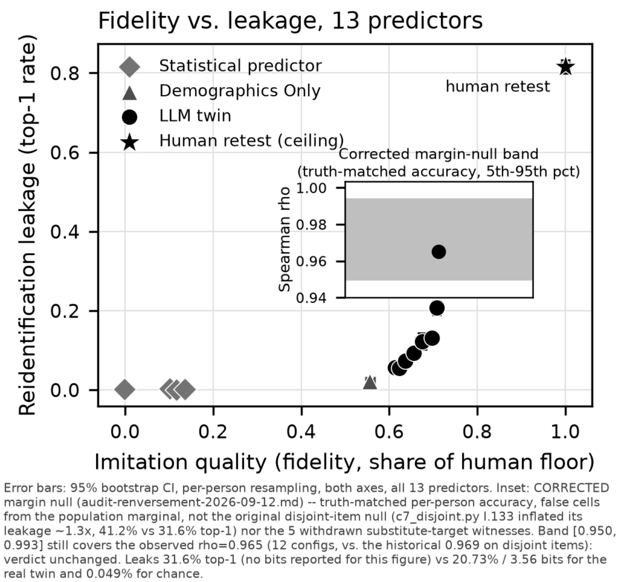

Figure 2: Imitation quality against leakage, with its control. Abscissa: individual fidelity (`fidelite_plancher`, share of the human floor, 0 to 1, linear). Ordinate: closed-world top-1 leakage, linear, 0 to 0.85. Thirteen points: the LLM twins (filled circles), the statistical reference points (grey diamonds), Demographics Only broken out (triangle), and the human retest (star) at fidelity ≈ 1 / leakage 81.6 %, all with error bars on both axes from a per-person bootstrap of 2,000 resamples. Inset: the 5th–95th percentile envelope of the corrected marginal null (mean rho 0.974) as a grey band, with the observed rho (0.965) plotted inside it. *At a glance*: the points rise together from left to right with no plateau at the top, but the observed correlation sits inside the band that a null matched only on accuracy already produces — the coupling is measured, not shown to be individual-specific. Data: `resultats/c7-compromis.csv`, `resultats/c7-compromis-robustesse-points.csv`, `resultats/c7-disjoint-nul.csv`.

6 Defense

Shuffling the purchase answers between people of the same demographic segment (D4) brings closed-world top-1 from **20.7 % [19.0 ; 22.4]** to **0.13 % [0.01 ; 0.28]**, per-item distribution and between-segment differences being preserved **exactly, by construction**, as PRAM predicts.

The cost is not a single average. Reported by component: 0.0 points on the per-item distribution, 0.0 points on group differences, and **4.4 points on inter-item correlations**. We therefore withdraw the summary that presented D4's cost *as* a single mean of 1.47 points: that figure is the unweighted mean of the three components, two of which are zero by construction, so quoting it alone divides by three an effect that falls entirely on the third. The quantity itself survives as a **composite utility index** — the one scale on which the comparison with differential privacy can be made at all (§ 6.2)

— and is named as the index wherever it appears below, never as the cost of the defense.

The cost is worse than that summary suggests. Measured against the real human answers, `erreur_correlations_hum` rises from **5.775 to 9.709** after D4, a degradation of **68.1 %**: the defended twin moves *away* from the humans on correlations. Any framing in which the defense's cost is absorbed by an error already present compares an increment to a level, and is incorrect.

Alternatives: D1 (k=10 aggregation) gives 0.55 % for 3.8 points across all three components; D2 (noise) never descends below 1 %.

6.1 The defense against a strong, then an adaptive attacker

The defense figure above was measured against the naive attack, which § 5.8 shows can badly understate leakage. We therefore reran it against the strong attack recalibrated on the *defended* outputs, then against an attacker who knows the mechanism and which items it leaves untouched. D4 holds. Top-1 goes from 0.13 % (naive) to **0.24 %** under the recalibrated strong attack, and the **adaptive** attacker plateaus at **0.29 %** (strategy S1, leaving the 20 opinion items intact; segment-invariant strategy S3 alone 0.05 %). That is two orders of magnitude below the 20.7 % undefended rate, and never above 1 %. **Attribute disclosure is not demonstrated either**: the best strategy names the correct `S_gra` segment in **7.7 %** of cases, against 6.8 % at chance and **12.6 %** for the trivial rule of always answering the most frequent segment — the attack does worse than not attacking.

A reservation that bounds the guarantee. This residue comes *entirely* from the 20 opinion items D4 does not permute. The guarantee holds for this split, not for a design leaving a more informative block untouched: 0.29 % is not a bound on the defense in general.

6.2 Differential privacy: our prediction refuted, and the argument that survives

We preregistered the prediction that at a moderate budget, differential privacy would be dominated by D4 on the aggregate risk-utility table. **It is not.** With per-item marginals perturbed by Laplace noise (L1 sensitivity 2, budget split over 60 items, basic sequential composition) followed by i.i.d. per-item sampling — degree-0 PrivBayes, no joint structure — the DP synthesiser at eps = 3 and eps = 10 loses **3.33** and **3.38** points on the **composite utility index**, within 5 points of D4's **1.47** on that same index, while its top-1 (**0.158 %**, **0.000 %**) is **not above** ours (0.13 %). The comparison runs on the aggregate index because that is the only scale on which the two mechanisms are commensurable; that index is not D4's cost, which falls entirely on inter-item correlations (§ 6). We do not claim our defense beats DP, on either axis.

The utility floor is architectural, not budgetary: the non-private control (eps = ∞) still loses 3.52 points, only distribution error falling with epsilon (9.7 ⟶ 3.2 points).

What survives is a theoretical point, not an empirical win. DP protects an individual's *membership* in the dataset used to compute a published statistic. Our attack assumes the person is already known, since her profile is the twin's input, and asks whether **the output conditioned on her** can be linked back: record linkage, not membership. The DP generator escapes that attack only by **never conditioning on an individual** — individual fidelity stays at most 0.2 points at every budget, including infinite — that is, by refusing the twin's task altogether. A panel holder who needs population statistics should use DP; one who needs a per-person twin cannot obtain one from this mechanism at any epsilon.

Limits. Written by hand rather than with a reference library, and not full PrivBayes; epsilon covers only the published histograms, with no composition across the repository's other analyses; correlations, between-segment differences and the `S_gra` covariate are not protected.

6.3 What the defense costs a downstream analyst

A defense is only usable if the analyses people actually run survive it. We measured three on the same 2,058 persons. **Preserved exactly**: group comparisons are **identical to 16 decimal places** between the raw and defended twin, and demographic regression coefficients keep their sign and significance. **Destroyed**: purchase-item coefficients significant in the raw twin become non-significant after D4, and on a PCA of the 40 purchase items **45 % of the first component's loadings are inverted** relative to humans, against **0 %** for the raw twin.

The honest comparison. The unprotected twin was already wrong: it inverts the sign of the male-female gap (+0.010 against −0.047 in humans) and already loses 5.3 points of PC1+PC2 variance. D4 adds **3.5 points**, less than the error already present — but the axis inversion is **D4's own doing**: it changes *which* items compose the structure more than its strength. With a D4-defended twin, group analyses and predominantly demographic regressions remain reliable; anything resting on the link between two answers of the same person becomes unusable.

Limits. Tested on a single block, of a single twin, of a single dataset.

7 Discussion and Limitations

7.1 The fourteen refuted preregistered predictions, two inconclusive, and one untestable

Rows 9 and 16 were listed as refutations until adversarial review found that a percentile bootstrap on a zero-event sample can only return "[0 ; 0]" — an arithmetic property of the resampling, not a confidence interval; recomputed exactly, both are inconclusive for lack of power. Row 17 left the count later still, for a different reason: its arithmetic is exact, but the pipeline that produced it carries no individual information (§ 5.4). Seventeen rows, fourteen refutations, two inconclusive, one untestable.

Three further preregistered outcomes were partial rather than refuted, and we count them as neither confirmations nor refutations: D4 erases two publishable inter-item effects while leaving signs unchanged, and its PCA prediction holds on variance but is contradicted on the loadings (§ 6.3); the paid frontier-model run cleared its fidelity > 0.10 bar at 0.1714, barely above our cheapest twin (§ 5.7). One prediction held outright under adversarial pressure: P2, that D4 stays below 1 % against a strong attacker (§ 6.1). Three verdicts reached after this table was closed are reported in

Table 1: The seventeen preregistered predictions and their outcomes (§7.1).

#	Prediction (preregistered)	Outcome	Source
1	The quality-leakage coupling exceeds a null matched on accuracy alone	**Refuted.** Null rho 0.974, 5th–95th percentiles [0.950 ; 0.993], against 0.965 observed over the 12 configurations; original null carried a defect, since corrected, verdict unchanged (§ 5.1)	`audit-renversement-2026-09-12.md`
2	Open-world > 5 % (Twin) and > 30 % (Park) at FPR = 1 %	**Refuted as first measured** (3.04 % and 20.39 %). Under the strong attack Twin still fails at 4.28 %, Park **passes** at 60.17 %	`c7-attaquant-fort-resultats.md`
3	Twin-to-twin top-1 ≥ 20 % on the Park archive	**Refuted.** 11.9–13.0 % at 177 common items (failure criterion not met either)	`c7-transfert-stanford-resultats.md`
4	The twin-to-twin channel leaks on the 19-common-item pairs	**Refuted.** 0.45 % mean top-1, at chance (0.05–0.10 %)	`c7-transfert-resultats.md`
5	H1: per-item entropy drives identification	**Refuted.** Opinion items 2.10 bits, identify 46× less	`c7-mecanisme-resultats.md`
6	H4: twins are more stereotyped than humans	**Refuted.** 36.2 % [35.6 ; 36.8] vs 37.5 % [36.9 ; 38.1]	`c7-mecanisme-resultats.md`
7	Deviations alone carry ≥ 80 % of the leakage	**Refuted.** 0.68 % in the mixed condition	`c7-deviations-resultats.md`
8	R1 ≥ 10 % on ≥ 2 of 3 of our own models	**Refuted.** No model reaches 1 %	`c7-gen-resultats.md`
9	Call granularity explains the leakage	**Inconclusive, underpowered.** 0.00 % both arms ([0 ; 8.8] at n=40, [0 ; 30.85] at n=10); top-10 runs opposite	`c7-recette-resultats.md`
10	Cost per unit of individual fidelity is roughly constant	**Refuted at equal sample.** CV 0.436 vs 0.357 on the same 9 points	`c7-compromis-resultats.md` § 5
11	Per-item entropy correlates with identifying power consistently	**Refuted.** Opposite sign by dataset: r = −0.81 (Twin), +0.57 (Park)	`c7-bits-resultats.md` § 3
12	P1: a stronger attacker gains ≥ 20 % relative over the naive attack	**Refuted on Twin** (+12.2 %, 20.7 → 23.23 %); held on Park (+38.0 %, 65.51 → 90.40 %)	`c7-attaquant-fort-resultats.md`
13	P3: an adaptive attacker knowing the mechanism breaks the defense	**Refuted, in the defense's favour.** Plateaus at 0.29 %, never above 1 %	`c7-attaquant-fort-resultats.md`
14	At a moderate budget, DP is dominated by our defense on the aggregate table	**Refuted.** eps = 3/10 lose 3.33/3.38 points on the composite utility index against D4's 1.47 on that same index (never D4's cost, § 6); top-1 0.158 % / 0.000 % against 0.13 %	`c7-dp-resultats.md`
15	A frontier model on Twin's per-item recipe reaches accuracy > 0.55	**Refuted.** 0.4722 [0.4361 ; 0.5050]	`c7-fort-resultats.md`
16	That same twin reaches top-1 > 5 %	**Inconclusive.** 0.00 % [0 ; 11.57] (Clopper-Pearson, n = 30), the 5 % threshold lying inside the interval	`c7-fort-resultats.md`
17	T2, independent pipelines: top-1 CI excludes the segment control and stays ≥ 2× the demographic baseline	**Not testable with this instrument — counted as neither.** 1.8 % [0.4 ; 3.6] at n = 142/200, but both arms' twins identify the real person at chance, so the contrast decides nothing either way (§ 5.4)	`c7-deux-organisations-resultats.md`

place and not counted in it: the distance law is **refuted** on its own terms (§ 5.4), and the third dataset's two predictions are **missed but out of domain** (§ 5.8).

7.2 What we do not know

An excess cost in bits per unit of fidelity, real but unexplained. On equal samples, CV(bits/fidelity) = 0.436 against CV(bits/accuracy point) = 0.357 over the same 9 points. With so few points and no identified mechanism, the excess **must not be attributed to "LLMs" as a family**.

We did not reproduce the leakage with our own twins (§ 5.7), including with a paid frontier model on Twin's own per-item recipe — an open limit, never a mechanism, and a narrower one than before: neither the model nor the call granularity, leaving the persona's format and length.

Rates do not predict across datasets. The normal-maxima model lands close on Park k = 60 (0.354 predicted, 0.340 observed) but fails both its controls (0.053 for 0.206 observed; 0.983 for 0.656): an isolated success between two failures is a coincidence, not a law. Only conditional information retains sign and order of magnitude across datasets (r = 0.72 and 0.89). The twin-to-twin channel's cross-dataset rate is likewise only partly explained, a residue of roughly 3× surviving both controls (§ 5.4). And nothing we hold separates the trajectory of § 5.8 from the protocols that differ along it (§ 1.1).

7.3 Multiplicity

Most of this paper's central results are bootstrap confidence intervals around a descriptive measure — re-identification rate, rank correlation, bits of identity — not classical hypothesis tests; of the 34 preregistered predictions underpinning those claims at the time of the census, 11 were refuted and 3 judged inconclusive, roughly one third, the opposite of the signature of data dredging. Of the 3 tests carrying a classical p-value, a Holm correction leaves the two smallest standing (0.002 and 0.008, adjusted to 0.006 and 0.016) and confirms the non-significance of the third; the intervals carrying the other claims must be read as measurements, each with its own margin, not as independent rejections of a common null.

The full census covers 47 adjudicated tests across three families — the article's claims, the standalone controls, and the abandoned branches — with 25 confirmed and 15 refuted. It was written over the 15 `c7-*` sub-studies existing at the time and does not include the verdicts added later (the strong attacker, DP, downstream utility, the paid frontier-model run, the distance law and the third dataset), which is why Table 1 and the census do not carry the same totals.

A second pass extended the confirmatory family to 15 tests convertible to a p-value and applied both Holm and Benjamini-Hochberg. **Neither changes any verdict**, and the two agree on all 15: the p-values are bimodal — eight at or below 0.0025, seven already near 1 — so no claim here rests on a marginally significant result. It also flags three thresholds as fragile, which we name rather than let a reviewer find: the frontier twin's **fidelity > 0.10** bar, confirmed at 0.1714 but refuted had it been set at 0.18 — while our own weakest known twin sits at 0.177; the choice of k in top-k (§ 5.2); and the functional form of the scale law beyond the measured range (§ 5.9). A common bootstrap across analyses reusing the same 2,058 people was abandoned for cost, so the intervals of § 5.1–§ 5.3 on Twin carry no simultaneous coverage.

7.4 Scope, and two limits established by failure

Three datasets, one language, one questionnaire format, categorical closed-choice answers only. The mechanism analysis and the defense both rest on a single block of a single dataset. We make no claim about conversational agents or panels outside the three studied here. Two further limits we state with the specific resource that would lift them, because we looked for that resource and did not find it.

Generality beyond American attitude instruments, and the audit bottleneck. All three datasets are American survey panels built on neighbouring instrument families, so we cannot say whether this channel is a property of that family — nor, as § 1.1 states, separate the trajectory from the protocols that differ along it. The obstacle is not that the method is rare, but that the matched file — row-by-row, not by demographic cell — is almost never redistributed. We verified three such datasets at the source; we ran no systematic census, so no upper bound is established. **Not a count, a typology**: two teams withhold their file (editorial choice or silence); a third is barred by its source licence (SOEP); and one inverse case exists, synthetic outputs with no matched humans. **The risk is created faster than it becomes auditable.** The resource that would lift our limit is a matched individual dataset from a domain distinct from GSS and Big Five for which twins produced by a **third-party team** already exist for the same respondents, under a licence permitting local computation and publication of an aggregate rate; and, for the trajectory of § 5.8, several model generations over the same items and the same population.

Free-text outputs. We measure nothing about free text, because no dataset here permits it: the Park archive publishes **no transcripts**, **none of the 13 Twin configurations** produces text, and the Argyle twins are single recoded tokens. Every rate in this paper therefore concerns closed-choice categorical answers, and we extrapolate them to free-text outputs in neither direction — the free-text inference literature [23, 35] attacks a richer signal than ours, so our figures are not an upper bound there. The missing resource is Park's complete transcripts, or a run of the Twin pipeline that generates text.

Ethical Considerations

Datasets, and the terms we are bound by. Twin-2K-500 is public under CC BY 4.0 and the Argyle archive under CC0 1.0. The Park replication package is publicly posted on OSF (node `t6g7k`), but public posting is not a licence: the OSF API returns `node_license: null` for that node, so **no licence is declared** and public access carries no right to redistribute the individual responses it contains. Its questionnaire content is a General Social Survey instrument, and **NORC's terms of use forbid reproducing the GSS in any form without prior written consent**. We record these constraints because an article about privacy that passes over the terms of use of its own data disqualifies itself.

What we do in consequence, verified rather than assumed. We redistribute no individual response file from any of the three archives. We checked our own released material file by file: the review artifact (Open Science) ships only code and a synthetic dataset generated at run time and reads nothing from `data/`, which is excluded from version control; every published table under `resultats/` is an aggregate by condition, by item or by bootstrap replicate; and the Argyle analysis prints and writes no ANES case identifier and no individual match. Only analysis code and disclosure-reviewed aggregates are published, with the NORC citation and attribution to Park et al. Two limits we state rather than resolve: NORC's published terms address reproduction of the GSS and do not speak to derived aggregate statistics, the reading that per-item aggregates fall outside the prohibition being ours and not NORC's; and the ANES clause covering the public variables the Argyle archive carries was not verified at its own source, the Dataverse deposit itself being CC0.

No new data was collected: the attack uses only already-published content (human answers and twin outputs). A limit to note: participants did not specifically consent to a re-identification test on the twins generated from their answers.

No individual is identified. No identifier linking a record to a real person is published or listed; only rates aggregated over the full cohorts are reported.

No new harm on Twin-2K-500. It already publishes, for each person, their identifiers and real answers beside their twin: matching is trivial without any attack. The result documents a generic risk for anyone releasing "anonymised" twins without that key — a practice Twin-2K-500 does not follow.

The Park archive is a different case, and we make no such claim there. That argument rests on a key published beside the twins and does not transfer: no trivial alignment makes the matching free there, the rates are far higher, and the objects we attack are the very ones whose individual release the authors treated as a privacy matter. What limits the harm is narrower: we publish no individual record and no per-person result from that archive, only aggregate rates, and the disclosure below is addressed to its authors, whose own warning this work quantifies.

Benefit. Several teams publish individualised LLM twin outputs without any linkage risk analysis. This provides a first measured figure for that risk on public datasets, and a reproducible method — attack plus interpretability control — for testing it *before* publication.

Responsible disclosure, and its actual state. Letters to the Twin-2K-500 authors and to Park et al. are drafted, with a 30-day response window and an offer to share code and report in advance. The tone toward Park et al. is fixed: *your warning was accurate, here is its measured magnitude.* **As of this submission the letters have not been sent and the 30-day window has not opened.** Sending them is on the critical path and is to happen before any preprint posting or code release. We write this plainly rather than imply a consultation that has not taken place: the committee should know the window was still closed when this manuscript was filed.

The Park et al. case. Their supplementary material already warned participants that their information might be "inadvertently shared" and acknowledged that complete anonymity remains challenging. They named the risk, obtained an ethics agreement worked over more than six months, pseudonymised, restricted access to part of the material and planned a 25-year withdrawal. What is nevertheless public is the replication package we analyse, which carries the 1,052 participants' individual responses together with the agent conditions' responses (§ 4.1): the protection announced in the paper does not cover the objects this attack consumes. We flag that gap as a question for the authors rather than a finding against them — from outside we cannot tell whether the restriction was lifted or never covered the replication package — and it is precisely what the disclosure letter must state correctly. No public audit had measured the magnitude of that named risk; our figures (90.40 % closed-world, 60.17 % open-world at 1 % false accusations, and 44.7 % from the interview alone, a lower bound) are, to our knowledge, its first quantification, and our own first measurement understated it.

Publishing the attack code. *For*: replicability, and enabling other panel holders to test their twins before publication. *Against*: it lowers the cost for an attacker already holding a third-party panel's real answers and a corresponding twin file. *Our resolution*: publish, because the code only acts if the attacker already holds real answers to the same items, and the countermeasure is straightforward once the risk is known.

Recommended defenses. (a) Do not publish twin outputs item by item for blocks with high inter-person specificity; publish aggregates or calibrated noise. (b) Break the person-twin alignment in public files. (c) Publish regenerated twins without copying wave metadata (`StartDate`/`EndDate`/`Duration`/`RecordedDate`). (d) Run a minimal linkage test, and the interpretability control of § 4.5, before any twin release: both are free, and the second says whether the first can be interpreted at all. (e) Within-segment shuffling (§ 6), with its real cost stated: 4.4 points on inter-item correlations, a 68.1 % worsening of the gap to human correlations (5.775 → 9.709), and the loss of every inter-item analysis downstream (§ 6.3) — it holds at 0.29 % against an attacker who knows the mechanism, but only for a split that shuffles the informative block (§ 6.1). (f) Differential privacy where the published object can be a population statistic rather than a per-person twin (§ 6.2).

Public interest. The AAPOR report of 8 May 2026 [32] ranks synthetic response generation as the most risky of the core tasks it evaluates and names re-identification by linkage without quantifying it. A risk named by professional bodies but never measured justifies an independent measurement followed by responsible disclosure rather than silence.

Ethics review and scope. This study was not submitted to an external ethics panel such as an IRB. It relies exclusively on secondary data already made public by their original publishing teams, and no new data was collected from any person: we surveyed no one, interviewed no one, and had no contact with any data subject. On that basis we judged the study out of scope for human-subjects review, with the limit recorded above — participants did not specifically consent to a re-identification test on the twins generated from their answers. The scoping does not extend to the released artifact, which processes no real person's data at all (Open Science).

Open Science

Review artifact. `artefact/` replays the re-identification attack and its D4 defense in a single command (`./artefact/run.sh`), **with no real data and no real person**, in about 25 seconds on a laptop, without GPU, network, or any language model call. It runs on a fictional dataset of 600 synthetic persons at fixed seed with a tunable individual-signal dial, structurally comparable to the real data (categorical items, product × price matrix block, demographic segments, retest) but with different sample sizes and signal strength. It demonstrates that the two mechanics are the ones running in the paper: `attaque.py` and `defense.py` import their functions directly from `analyses/c7_reidentification.py` and `analyses/c7_defense.py`, nothing is copied, the only function that reads real data is never called, and a guard refuses to start if a path to `data/` is detected.

No artifact figure is a result of this study. The artifact's outputs (on fictional data: 21.9 % before defense, 0.842 % after D4) illustrate the mechanics and must never be cited as study results.

Underlying analyses. Every quantitative claim in this paper corresponds to a dated analysis script and result file under `resultats/` in this repository, organised to mirror the paper's own section numbering; individual file names are not cited in the text, to keep the manuscript readable.

Obtaining the real data. Twin-2K-500: Hugging Face repository `LLM-Digital-Twin/Twin-2K-500`, CC BY 4.0. Argyle et al.: Harvard Dataverse `doi:10.7910/DVN/JPV20K`, CC0 1.0. Park replication archive: OSF `https://osf.io/t6g7k/`, which carries real individual responses, is not redistributed here, and is never read by the artifact — and whose two obstacles public availability does not lift: no declared licence, and NORC's terms on the GSS content (Ethical Considerations). Obtain all three from their original sources and verify current terms of use before any download.

Preregistrations. `resultats/c7-preenregistrement.md`, `resultats/c7-defense-preenregistrement.md`, `resultats/c7-stanford-preenregistrement.md`, `resultats/c7-argyle-preenregistrement.md` and `resultats/c7-loi-distance-preenregistrement.md`, each written before any computation. Third-party timestamping is pending.

Environment. Python 3.13.14, numpy 2.5.2, pandas 3.0.5, scikit-learn 1.9.0. No network dependency.

AI Use

Large-language-model-based agents were used throughout this project: they wrote the analysis code, ran the experiments, drafted this manuscript, and conducted adversarial review of both the code and the text. We state this in full rather than narrowly, because an account that understated that involvement would mislead a reader more than a plain one. The responsible investigator directed the work throughout: set the research questions and the falsifiable predictions in advance of each analysis, decided what to keep, discard, or redo, authorized all paid computation, and is responsible for the accuracy and integrity of the submission. No generative AI tool is listed as an author.

Three checks bear on how much trust the model-produced outputs warrant. First, quantitative predictions were preregistered in writing, dated, before the corresponding analysis script was run. Second, the analysis pipeline fixes a random seed at every stochastic step, and independent reruns are checked to reproduce identical output before a result is reported. Third, the manuscript and code were subjected to adversarial review passes whose default posture is rejection; these passes are how two substantive defects were found and corrected before submission: a statistical control meant to carry only each person's accuracy margin that in fact copied their true individual answer with the complementary probability, and a bootstrap confidence interval reported as "[0, 0]" on a zero-event sample, since replaced with an exact Clopper-Pearson interval. These checks reduce, but do not remove, the risk that a model-produced number or claim is wrong; the responsible investigator remains accountable for every figure and statement in this paper regardless of how it was produced.

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